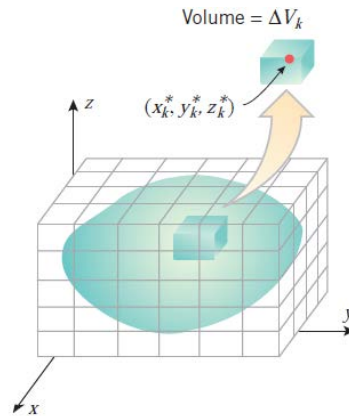


## 14.5 TRIPLE INTEGRALS

### DEFINITION OF A TRIPLE INTEGRAL

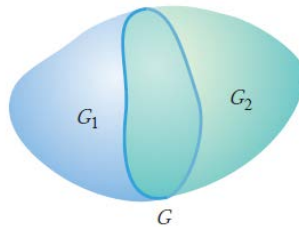
A single integral of a function  $f(x)$  is defined over a finite closed interval on the  $x$ -axis, and a double integral of a function  $f(x, y)$  is defined over a finite closed region  $R$  in the  $xy$ -plane. Our first goal in this section is to define what is meant by a *triple integral* of  $f(x, y, z)$  over a closed solid region  $G$  in an  $xyz$ -coordinate system. To ensure that  $G$  does not extend indefinitely in some direction, we will assume that it can be enclosed in a suitably large box whose sides are parallel to the coordinate planes (Figure 14.5.1). In this case we say that  $G$  is a *finite solid*.



▲ Figure 14.5.1

Moreover, if the region  $G$  is subdivided into two subregions  $G_1$  and  $G_2$  (Figure 14.5.2), then

$$\iiint_G f(x, y, z) \, dV = \iiint_{G_1} f(x, y, z) \, dV + \iiint_{G_2} f(x, y, z) \, dV$$



▲ Figure 14.5.2

**14.5.1 THEOREM (Fubini's Theorem\*)** Let  $G$  be the rectangular box defined by the inequalities

$$a \leq x \leq b, \quad c \leq y \leq d, \quad k \leq z \leq l$$

If  $f$  is continuous on the region  $G$ , then

$$\iiint_G f(x, y, z) dV = \int_a^b \int_c^d \int_k^l f(x, y, z) dz dy dx \quad (2)$$

Moreover, the iterated integral on the right can be replaced with any of the five other iterated integrals that result by altering the order of integration.

Six orders of integration are possible for the iterated integral in Theorem 14.5.1:

$$\begin{array}{lll} dx dy dz, & dy dz dx, & dz dx dy \\ dx dz dy, & dz dy dx, & dy dx dz \end{array}$$

► **Example 1** Evaluate the triple integral

$$\iiint_G 12xy^2z^3 dV$$

over the rectangular box  $G$  defined by the inequalities  $-1 \leq x \leq 2, 0 \leq y \leq 3, 0 \leq z \leq 2$ .

**Solution.** Of the six possible iterated integrals we might use, we will choose the one in (2). Thus, we will first integrate with respect to  $z$ , holding  $x$  and  $y$  fixed, then with respect to  $y$ , holding  $x$  fixed, and finally with respect to  $x$ .

$$\begin{aligned} \iiint_G 12xy^2z^3 dV &= \int_{-1}^2 \int_0^3 \int_0^2 12xy^2z^3 dz dy dx \\ &= \int_{-1}^2 \int_0^3 [3xy^2z^4]_{z=0}^2 dy dx = \int_{-1}^2 \int_0^3 48xy^2 dy dx \\ &= \int_{-1}^2 [16xy^3]_{y=0}^3 dx = \int_{-1}^2 432x dx \\ &= 216x^2 \Big|_{-1}^2 = 648 \quad \blacktriangleleft \end{aligned}$$

**14.5.2 THEOREM** Let  $G$  be a simple  $xy$ -solid with upper surface  $z = g_2(x, y)$  and lower surface  $z = g_1(x, y)$ , and let  $R$  be the projection of  $G$  on the  $xy$ -plane. If  $f(x, y, z)$  is continuous on  $G$ , then

$$\iiint_G f(x, y, z) dV = \iint_R \left[ \int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz \right] dA \quad (3)$$

► **Example 2** Let  $G$  be the wedge in the first octant that is cut from the cylindrical solid  $y^2 + z^2 \leq 1$  by the planes  $y = x$  and  $x = 0$ . Evaluate

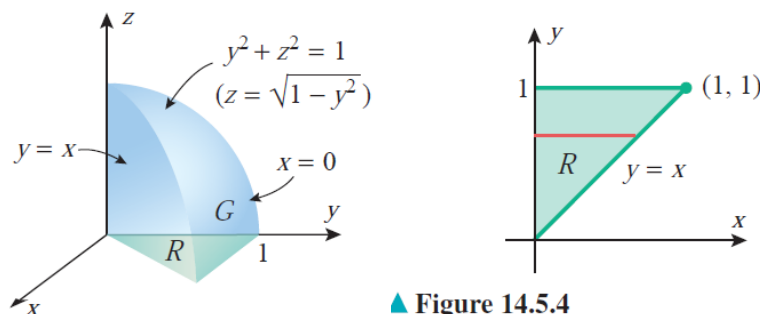
$$\iiint_G z \, dV$$

**Solution.** The solid  $G$  and its projection  $R$  on the  $xy$ -plane are shown in Figure 14.5.4. The upper surface of the solid is formed by the cylinder and the lower surface by the  $xy$ -plane. Since the portion of the cylinder  $y^2 + z^2 = 1$  that lies above the  $xy$ -plane has the equation  $z = \sqrt{1 - y^2}$ , and the  $xy$ -plane has the equation  $z = 0$ , it follows from (3) that

$$\iiint_G z \, dV = \iint_R \left[ \int_0^{\sqrt{1-y^2}} z \, dz \right] dA \quad (4)$$

For the double integral over  $R$ , the  $x$ - and  $y$ -integrations can be performed in either order, since  $R$  is both a type I and type II region. We will integrate with respect to  $x$  first. With this choice, (4) yields

$$\begin{aligned} \iiint_G z \, dV &= \int_0^1 \int_0^y \int_0^{\sqrt{1-y^2}} z \, dz \, dx \, dy = \int_0^1 \int_0^y \left. \frac{1}{2} z^2 \right|_{z=0}^{\sqrt{1-y^2}} dx \, dy \\ &= \int_0^1 \int_0^y \frac{1}{2} (1 - y^2) \, dx \, dy = \frac{1}{2} \int_0^1 (1 - y^2) x \Big|_{x=0}^y dy \\ &= \frac{1}{2} \int_0^1 (y - y^3) \, dy = \frac{1}{2} \left[ \frac{1}{2} y^2 - \frac{1}{4} y^4 \right]_0^1 = \frac{1}{8} \quad \blacktriangleleft \end{aligned}$$



▲ Figure 14.5.4

► **Example 3** Use a triple integral to find the volume of the solid within the cylinder  $x^2 + y^2 = 9$  and between the planes  $z = 1$  and  $x + z = 5$ .

**Solution.** The solid  $G$  and its projection  $R$  on the  $xy$ -plane are shown in Figure 14.5.5. The lower surface of the solid is the plane  $z = 1$  and the upper surface is the plane  $x + z = 5$  or, equivalently,  $z = 5 - x$ . Thus, from (3) and (5)

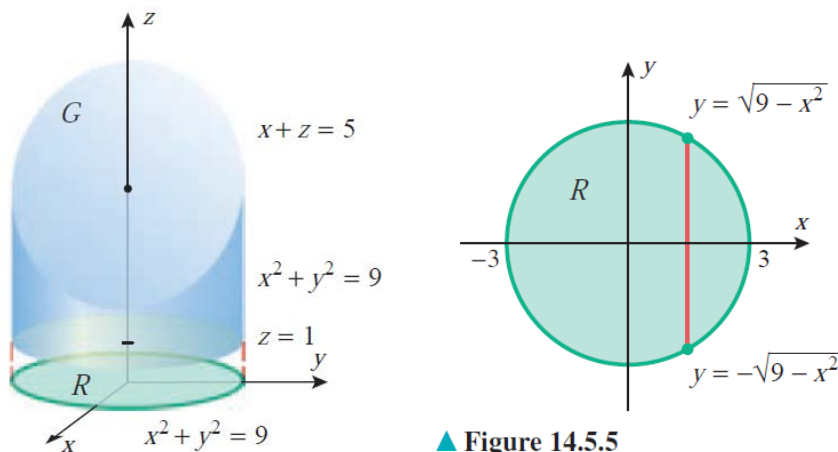
$$\text{volume of } G = \iiint_G dV = \iint_R \left[ \int_1^{5-x} dz \right] dA \quad (6)$$

For the double integral over  $R$ , we will integrate with respect to  $y$  first. Thus, (6) yields

$$\begin{aligned} \text{volume of } G &= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_1^{5-x} dz \, dy \, dx = \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} z \, dy \, dx \Big|_{z=1}^{z=5-x} \\ &= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x) \, dy \, dx = \int_{-3}^3 (8-2x)\sqrt{9-x^2} \, dx \\ &= 8 \int_{-3}^3 \sqrt{9-x^2} \, dx - \int_{-3}^3 2x\sqrt{9-x^2} \, dx \\ &= 8 \left( \frac{9}{2}\pi \right) - \int_{-3}^3 2x\sqrt{9-x^2} \, dx \\ &= 8 \left( \frac{9}{2}\pi \right) - 0 = 36\pi \quad \blacktriangleleft \end{aligned}$$

For the first integral, see Formula (3) of Section 7.4.

The second integral is 0 because the integrand is an odd function.



▲ Figure 14.5.5

► **Example 4** Find the volume of the solid enclosed between the paraboloids

$$z = 5x^2 + 5y^2 \quad \text{and} \quad z = 6 - 7x^2 - y^2$$

**Solution.** The solid  $G$  and its projection  $R$  on the  $xy$ -plane are shown in Figure 14.5.6. The projection  $R$  is obtained by solving the given equations simultaneously to determine where the paraboloids intersect. We obtain

$$5x^2 + 5y^2 = 6 - 7x^2 - y^2$$

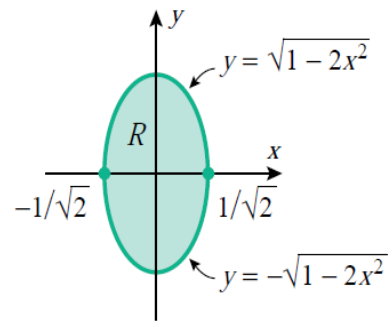
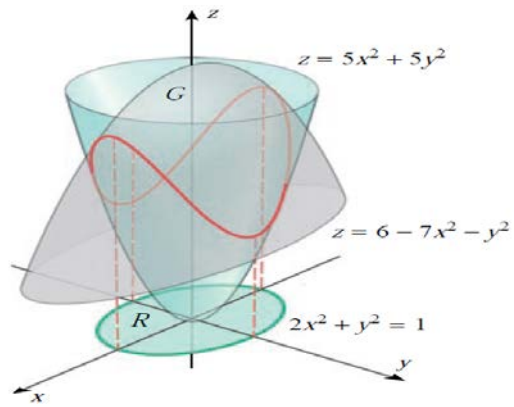
or

$$2x^2 + y^2 = 1 \tag{7}$$

which tells us that the paraboloids intersect in a curve on the elliptic cylinder given by (7).

The projection of this intersection on the  $xy$ -plane is an ellipse with this same equation. Therefore,

$$\begin{aligned} \text{volume of } G &= \iiint_G dV = \iint_R \left[ \int_{5x^2+5y^2}^{6-7x^2-y^2} dz \right] dA \\ &= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} \int_{5x^2+5y^2}^{6-7x^2-y^2} dz \, dy \, dx \\ &= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} (6 - 12x^2 - 6y^2) \, dy \, dx \\ &= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left[ 6(1 - 2x^2)y - 2y^3 \right]_{y=-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} dx \\ &= 8 \int_{-1/\sqrt{2}}^{1/\sqrt{2}} (1 - 2x^2)^{3/2} dx = \frac{8}{\sqrt{2}} \int_{-\pi/2}^{\pi/2} \cos^4 \theta \, d\theta = \frac{3\pi}{\sqrt{2}} \quad \blacktriangleleft \end{aligned}$$



▲ Figure 14.5.6

**Practice Problem:**

**Chapter-14.5:**

**Problems:** 5,6,8,10,11,15-18

**Thank you for your attention!**